PHYC10009 Physics Foundations

Power for civilisation
Week 12 Lecture 3

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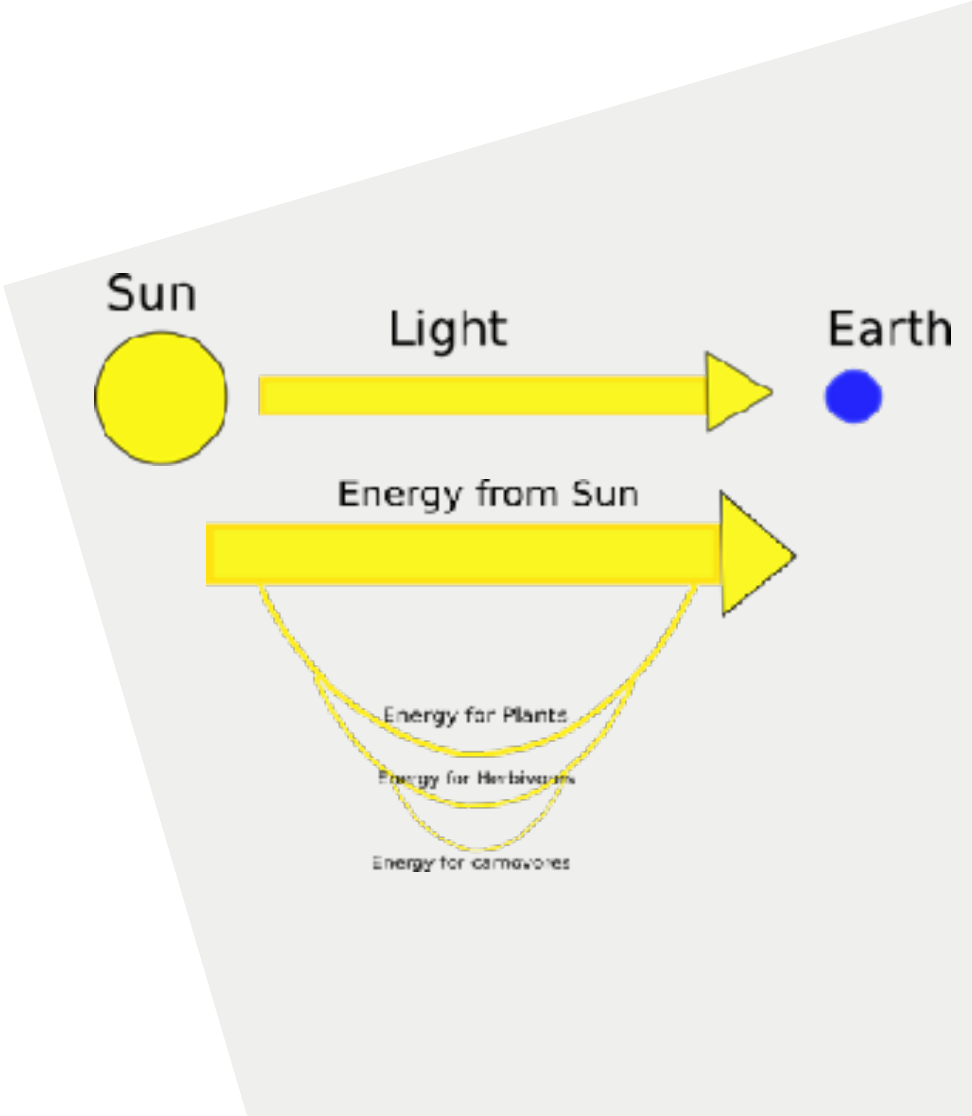
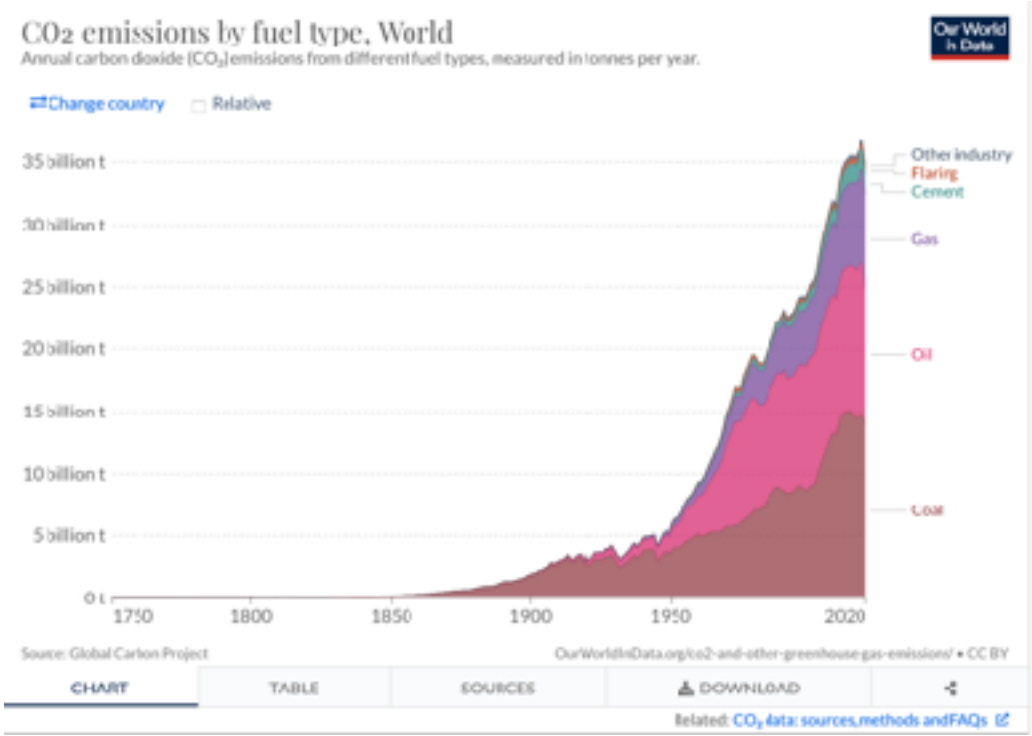
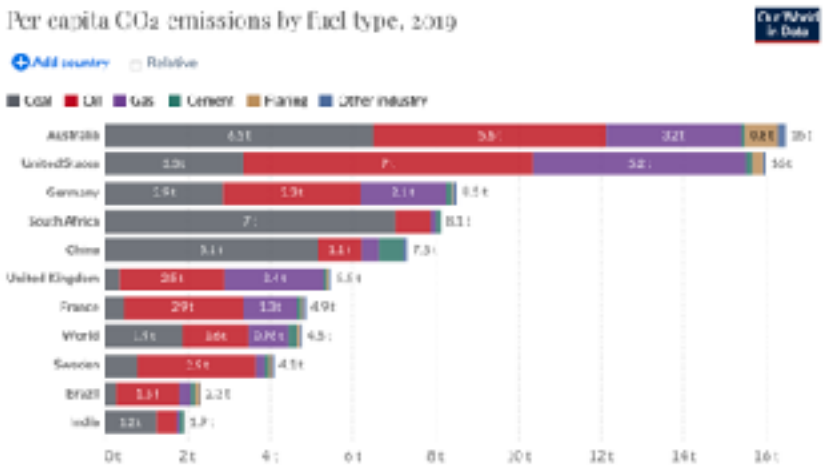


Power source for transportation - Oil

Liquid Hydrocarbons are both energy dense and easy to transport
These are the primary energy source for transport with non-pre-defined routes like cars, trucks, agricultural machinery, long distance trains, ships and aircraft.
They are almost all sourced from Crude Oil. This makes a really significant contribution to global CO_2 emissions.





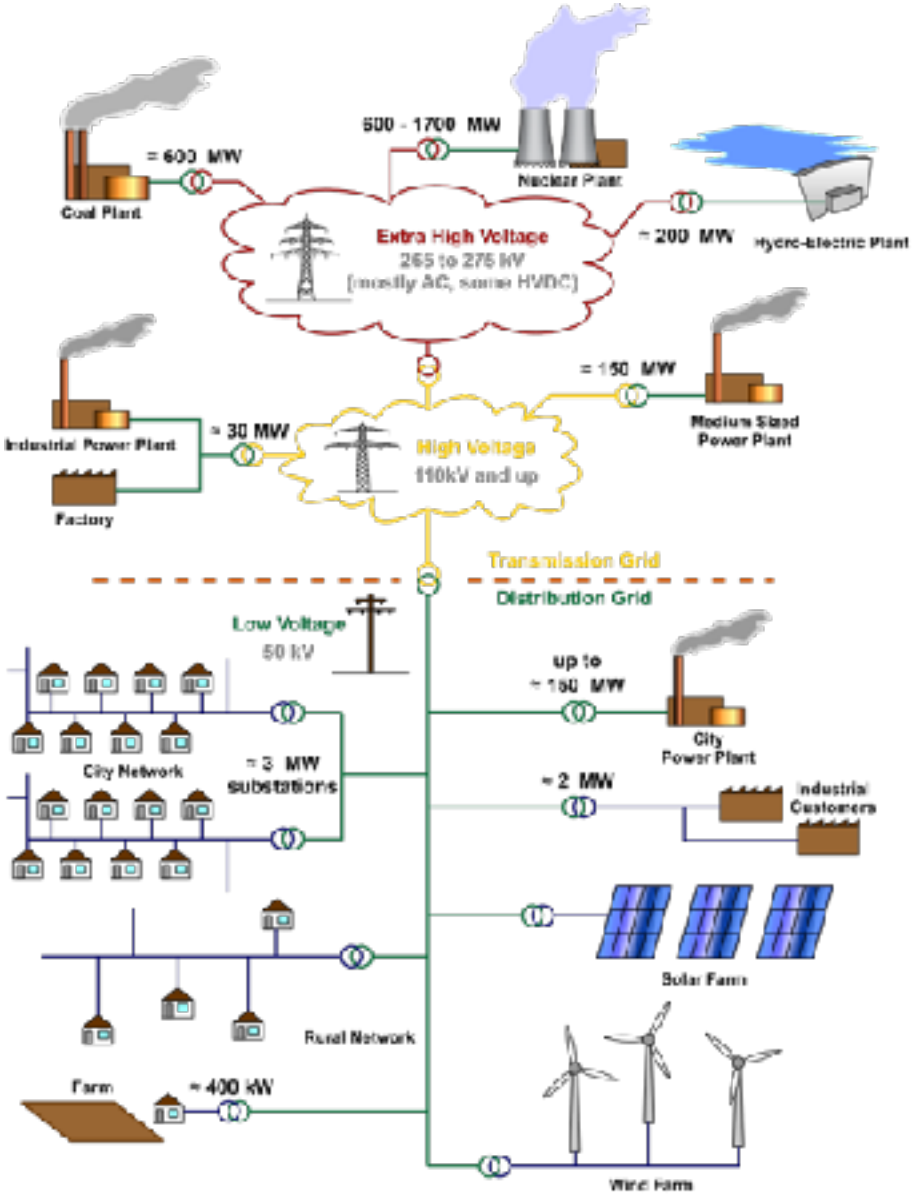
The versatility of Oil makes it difficult to replace

The electric grid

Electricity is an incredibly useful form of power which can be efficiently employed for many tasks.

Electrical power can be sourced via many different technologies and efficiently distributed where needed.

A typical Australian household consumes ~ 20 kWh of electricity per day



Transportation via Electricity

Trams and trains have fixed routes which enables large scale transportation via the electric grid.

By Zed Fitzhume - originally posted to Flickr as 755M looking flash in the new Metro livery at Flinders st., CC BY 2.0, https://commons.wikimedia.org/w/index.php?curid=8827356

Kevin Helon https://www.123rf.com/profile_khellon

They employ highly efficient electric motors and move many people at once.
Increased use of trains and trams will decrease CO_2 emissions by transport.
Limited scope for large scale reductions

Battery electric cars and trucks

- Employ advanced Lithium-ion batteries to store electricity to run electric motors.
- High efficiency of batteries, electric motors and regenerative braking compensate for the lower energy density of batteries.
- Currently higher purchase price than Internal Combustion vehicles
- Lower running costs partially balance this.
- Price declines are likely to push this technology to replace IC technology in the next decade

Electrification of automotive transport is vital for the "Electrification of Power" scenario

Power sources for electricity - Coal

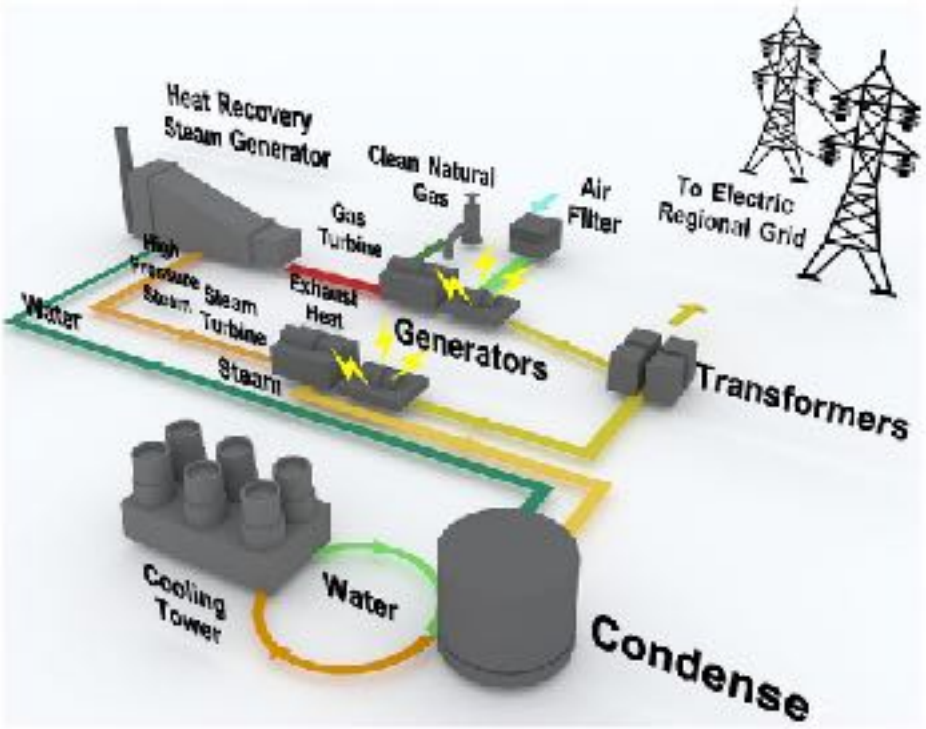
- Coal fired power stations burn coal to power Heat engines attached to electric generators.
- Typical efficiencies are 35%-45% with CO_2 emissions of 0.82 kg/kWh
- In addition to CO_2 emissions they make other contributions to air pollution.
- Cost depends on the number of pollution mitigation measures in place.
- Coal fired power can be cheap and nasty.
- Coal itself is relatively cheap but you need a lot of it.
- 1 GW station requires 9000 tonnes of coal per day for operation.
- It is “dispatchable” meaning if power is required at any time with sufficient notice the station can provide.
- Widely used as the fuel for the “first wave” of industrialisation of a country.

- **Needs to be severally curtailed to mitigate global warming.**

Power source for electricity – Natural Gas

- Natural gas is also employed to drive heat engines connected to generators.
- Because of the very high temperatures, the efficiency of “combined-cycle” plants can reach over 60%

- Less polluting than coal.
- CO_2 emissions 0.5 kg/kWh
- Cost of Natural Gas substantially larger than coal.
- Less efficient designs can be rapidly ramped up/down
- Can be employed to mitigate intermittency of Wind and Solar Power



https://www.coroflot.com/kevinjg6/combined-cycle-generation-process

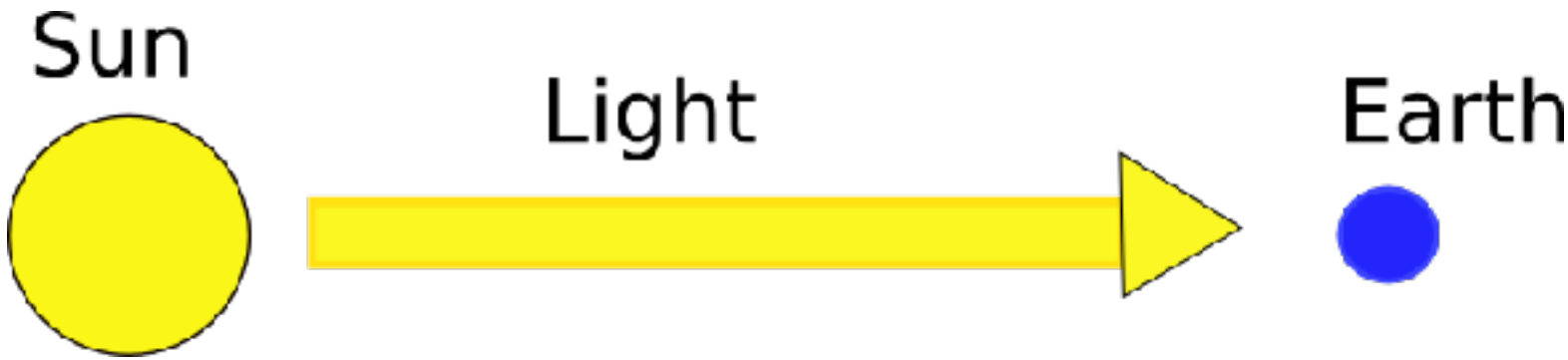
Natural Gas Power Plants will need to be phased out to meet CO_2 reduction targets

Low CO_2 Electric Power generation

Vital to Climate Change mitigation strategy is large-scale change to low CO_2 emission Electricity Generation technologies.

1. Nuclear Power (last lecture)
2. "Renewable Energy"

- Renewable Energy is ultimately resourced from the sun.



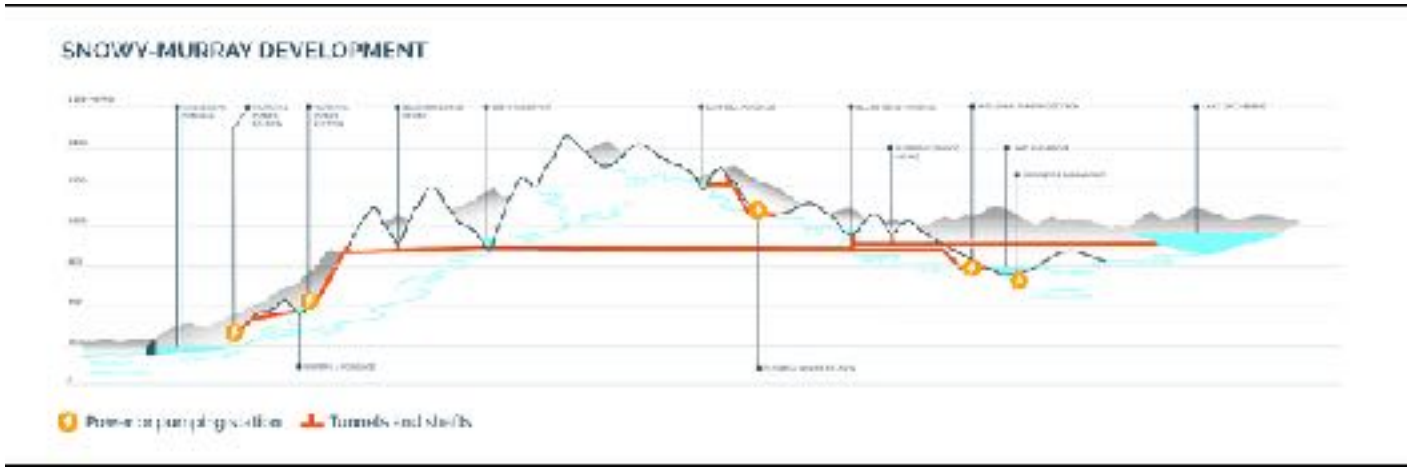
- Total Yearly Energy consumption of humans ~ 6×10^{20} Joules
- Yearly solar radiance on the Earth ~ 40,000 $\times 10^{20}$ Joules
- About 10000 times more than our needs
- Not all available but plenty of headroom!

Hydro-Electricity

Simple idea: Falling water is employed to drive mechanical generators

First large scale electricity generation technology. Very long lifetime.

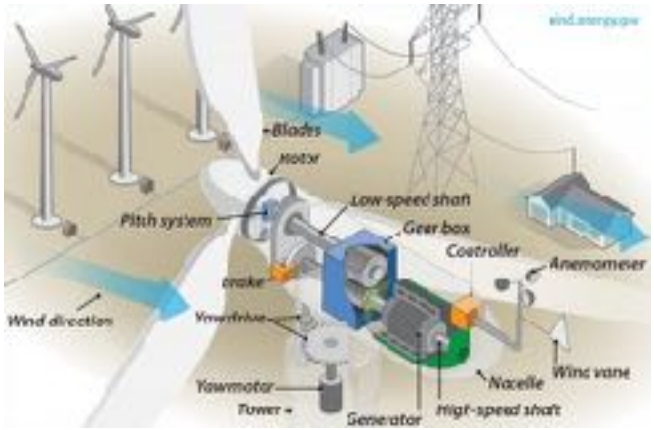
Normally need dams, tunnels and large capital works



Need a specific combination of weather and terrain to build.
Limited availability of such sites, particularly in Australia.
Very dispatchable provided sufficient water is available.

Wind Power

Employs wind to convert to mechanical power and drive electric generators
Wind for electric use is relatively recent.



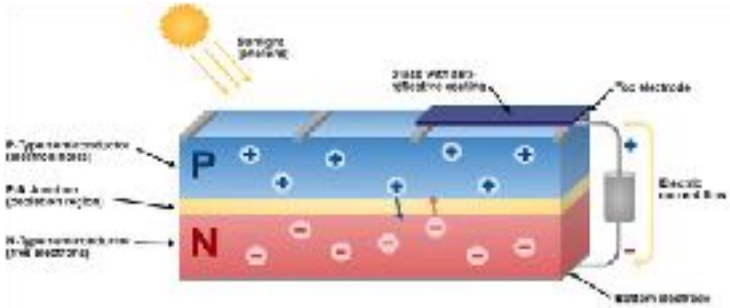
A modern wind turbine is a sophisticated collection of mechanical engineering.
Costs have reduced by over a factor 20 over the past 30 years

- Theoretical maximum efficiency is 55%
- Power is defined in terms of maximum output per turbine (2-15) MW
- Deployment depends on “capacity factor” of the location. Fraction of a year the maximum output is reached.
- A good location has > 30%, best ~45%
- “Dispatchability” is poor.

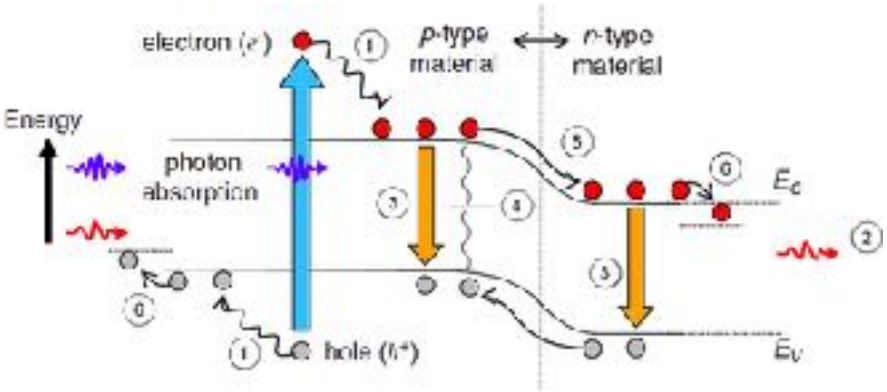
https://rameznaam.com/2014/09/29/the-renewable-energy-revolution/

Photovoltaic solar

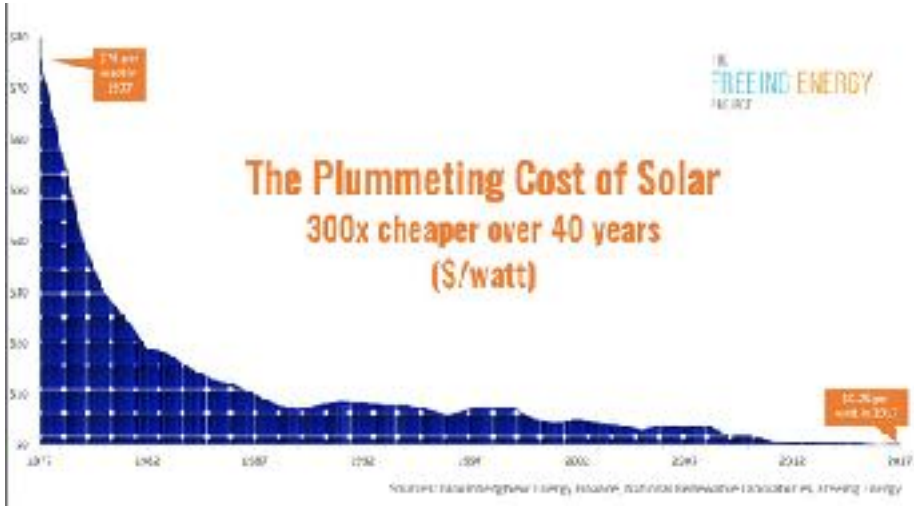
Directly employs principles of Quantum Mechanics to convert Solar Power to electricity.
Energy of the photons from light is directly converted to energy in an electric current



https://www.apricus.com/solar-pv-systems-pv-panels-19.html



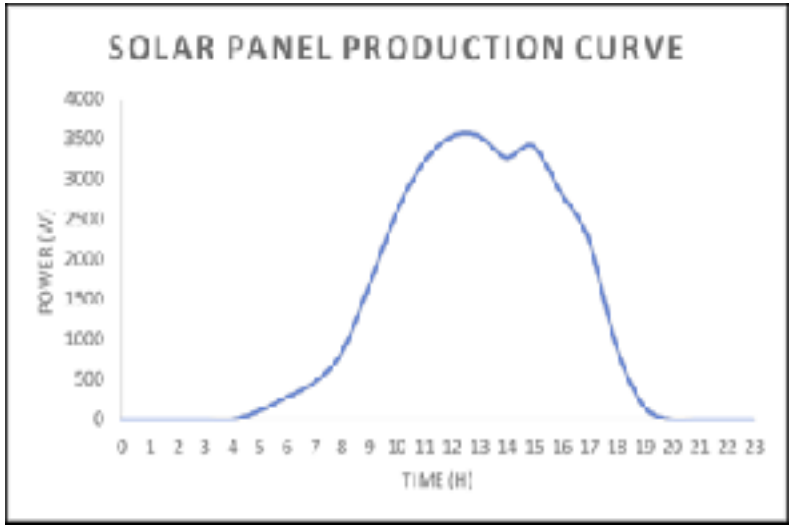
https://www.researchgate.net/figure/Model-of-a-p-n-junction-solar-cell-and-main-mechanisms-upon-absorption-of-photons_fig2_319646007



Current typical efficiencies ~20%
Reasonable extrapolations ~ 30% by 2030
Substantial price reductions likely to continue
Clearly only work during the day!

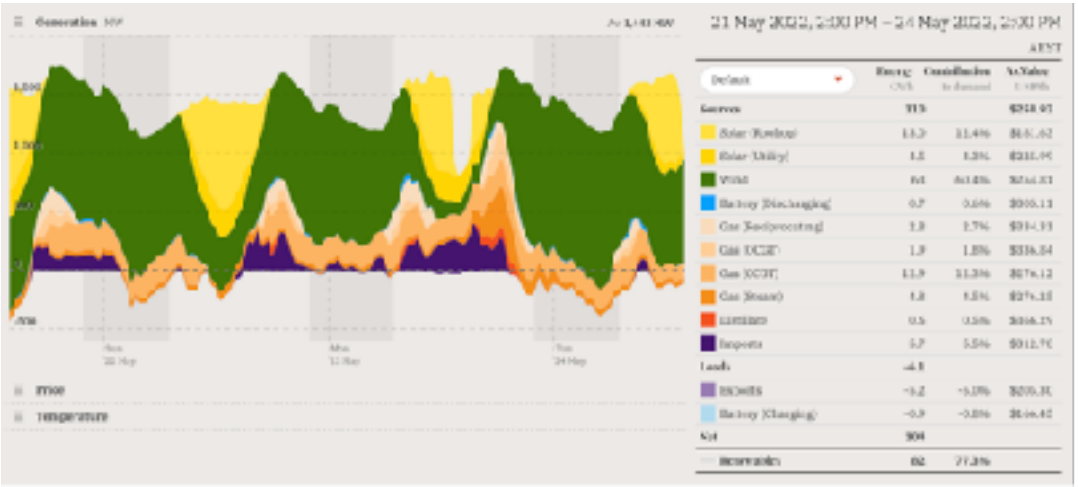
Solar insolation

- Solar power plants are rated by their maximum power output, when the sun is directly overhead and not blocked by clouds.
- The total output of the power plant is then rated by the locations' *insolation.*
- This is quoted as the fraction of the full 24 hours the plant will deliver it's maximum output. Good locations have an insolation > 25%



https://www.researchgate.net/publication/320827447_Mitigating_Brown-Outs_Fair_Distribution_of_Locally_Sourced_Energy_in_Smart_Grids

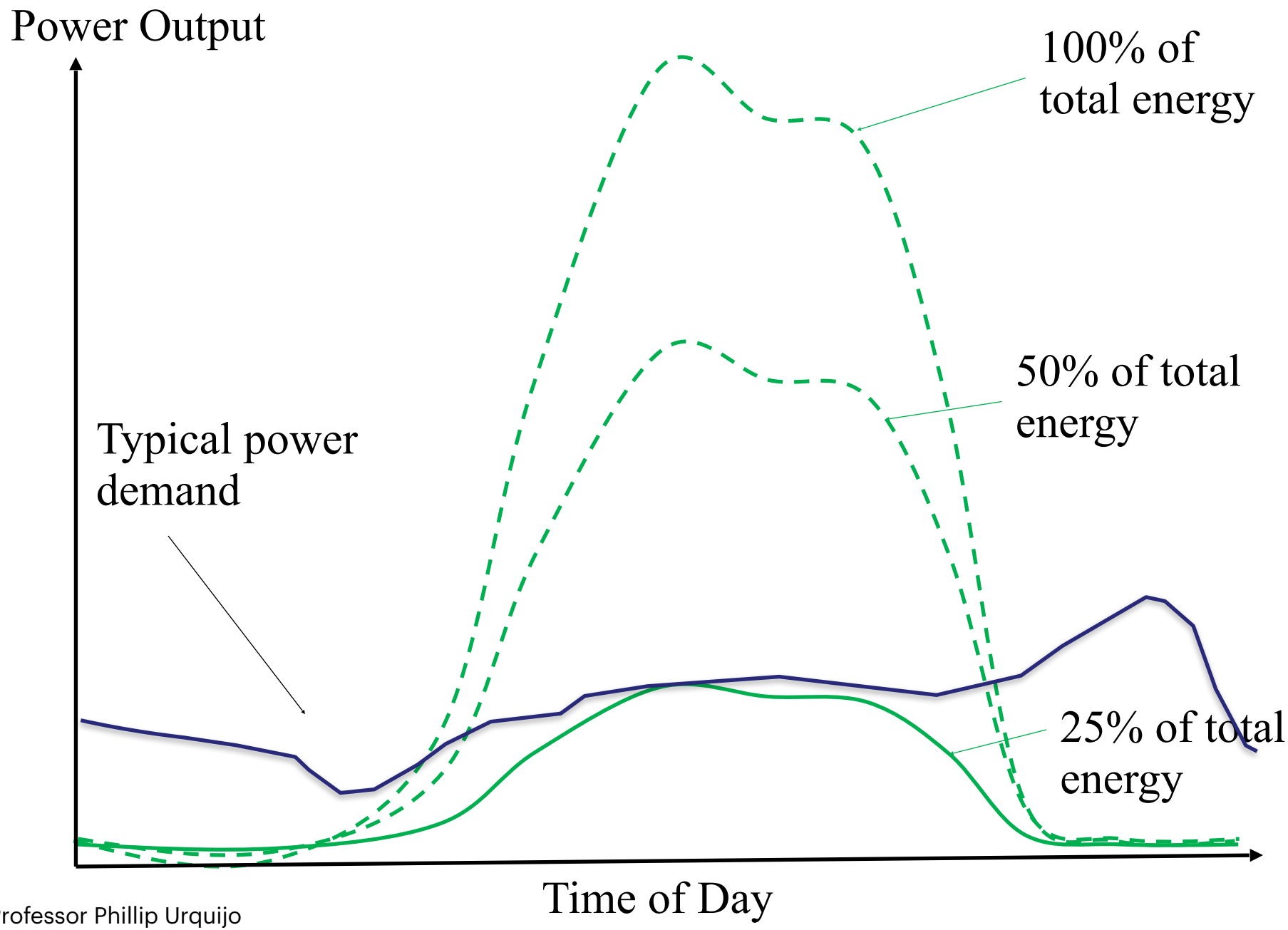
South Australian production, May 2022



https://opennem.org.au/energy/sa1/?range=3d&interval=30m

Solar PV will have to produced in excess of demand and stored

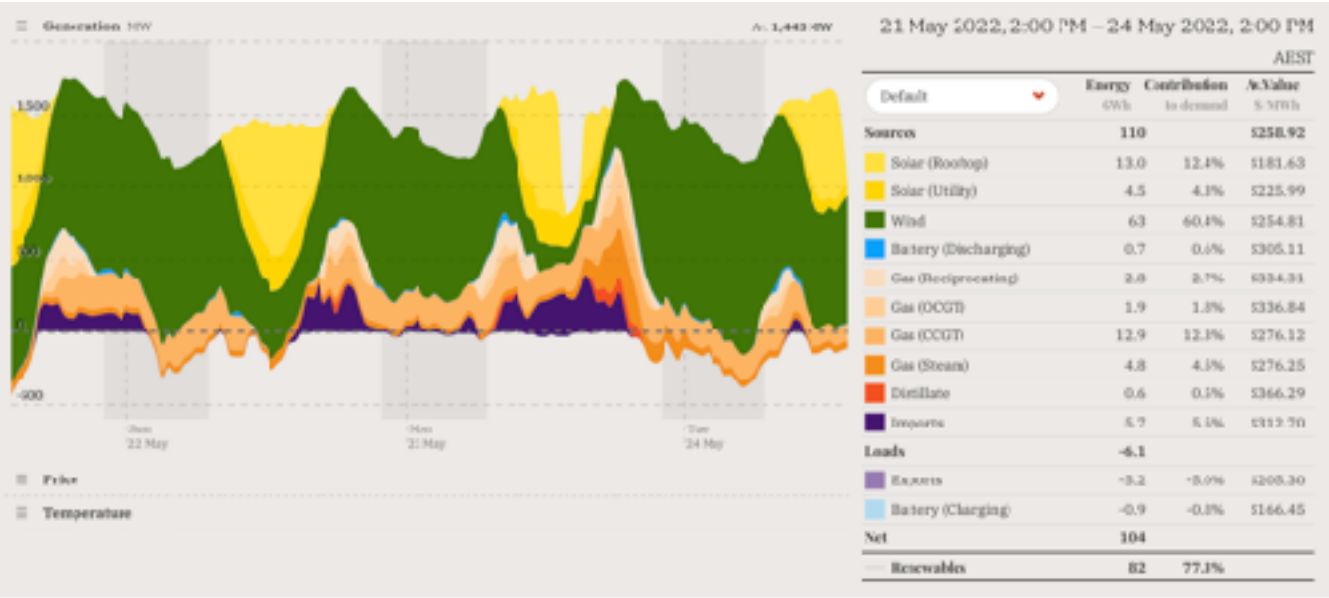
25% Insolation



Mitigating Renewable Energy intermittency

The two most scalable technologies for renewable energy production have significant issues with intermittency.
This becomes increasingly larger problem as these technologies make up a larger fraction of the grid.

Electricity Generation in South Australia



Total Renewable contribution to SA grid is about 75% in 2022.

To reach 100% there are times when power generation is too great.

Mitigating Renewable Energy intermittency

Fix with:
Improved Transmission (move local surplus to local deficits)
Grid-scale storage. (store excess energy and deliver as needed)
Example: Combination Snowy 2.0 scheme for SE Australia

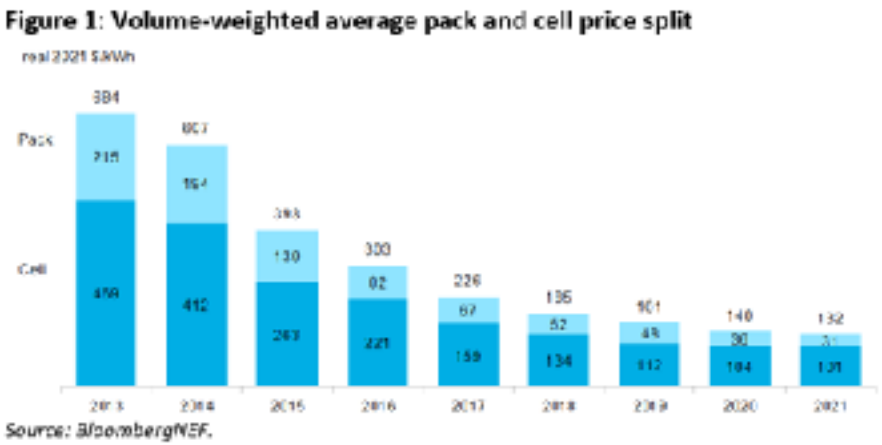
Collection and distribution network

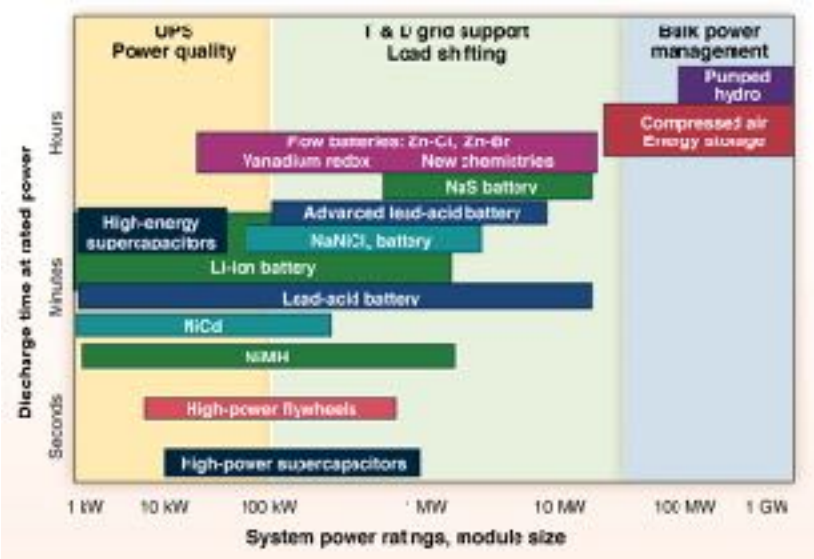
Pumped Hydro storage

Other possibilities include grid-scale Batteries
Incremental improvements continue to drive down costs

Lithium–ion Battery Prices

Incremental improvements in design and manufacturing drive cost reductions for Batteries.
Observed price declines are 15%-20% year.





Extrapolating to ~ 2025 has local grid scale storage costs as competitive with pumped Hydro.

Summary

- Our civilisation delivers a historically unprecedented level of comfort
- Ultimately this is a large local reduction of entropy for humans
- Requires very large-scale use of cheap power
- This is delivered via heat-engines and electricity
- Electricity enables highly efficient energy transformations
- Current source of energy, fossil fuels drives Global Climate Change
- We need to move to an even-more electrified civilisation
- Low CO_2 emission technologies are required for electricity production

Nuclear Power
Hydro Electricity
Wind Power
Solar Power
Renewable intermittency
Grid Power Storage

Review:

- You've learned a LOT about Science and how Nature works from this course.
- Here is a chance to review what we've learned by doing a collection of GAMSAT questions.
- Imagine trying to answer these without having done PHYC10009!

GAMSAT Question

Question 105. Making a few simple assumptions, which of the following is the best estimate of the rate at which your finger nails grow?

A $10^{-3}\ ms^{-1}$

B $10^{-6}\ ms^{-1}$

C $10^{-9}\ ms^{-1}$

D $10^{-12}\ ms^{-1}$

Cut my fingernails once every ~3 weeks when they're ~2 mm longer than minimum.

$$v = \frac{\Delta x}{\Delta t}$$

$$\Delta x = 2 \times 10^{-3} m\ ,\ \Delta t = 3 \times 7 \times 24 \times 3600 = 1.8 \times 10^{6}\ s \Rightarrow$$

$$v = \frac{2 \times 10^{-3}}{1.8 \times 10^{6}} = 1.3 \times 10^{-9}\ ms^{-1}$$ (Ans C)

GAMSAT Question

Questions 87-89

Humans have been broadcasting signals into space since the invention of radio transmission approximately 100 years ago. There is now an expanding three-dimensional sphere or 'bubble' of radio and television signals radiating out from Earth. Television broadcasts are of greater energy than the older radio broadcasts.

- 1 light year (the distance travelled by light in a year) is approximately 1×10^{16} m.
- The speed of light is approximately 3×10^{8} ms $^{-1}$.

87. Which of the following is closest to the diameter of the radio bubble?

A $1 \times 10^{18}\ m$

B $2 \times 10^{18}\ m$

C $3 \times 10^{24} m$

D $6 \times 10^{24}\ m$

Diameter $d = 2 \times v \times t = 2 \times 1 \times 10^{16} \times 1 \times 10^{2} = 2 \times 10^{18}\ m$

Ans B

88. It would be difficult for any extraterrestrial life to detect these signals because the signals lose intensity at a rate inversely proportional to:

A the radius of the radio bubble

B twice the radius of the radio bubble

C the radius of the radio bubble squared

D the radius of the bubble cubed

89. Two signals (**P** and **Q**) of the same frequency are transmitted from Earth. **Q** has 100 times as much energy as **P**. Suppose there is a minimum intensity at which signals can be detected. **Q** can be detected up to:

A the same distance as **P**.

B 10 times as far as **P**.

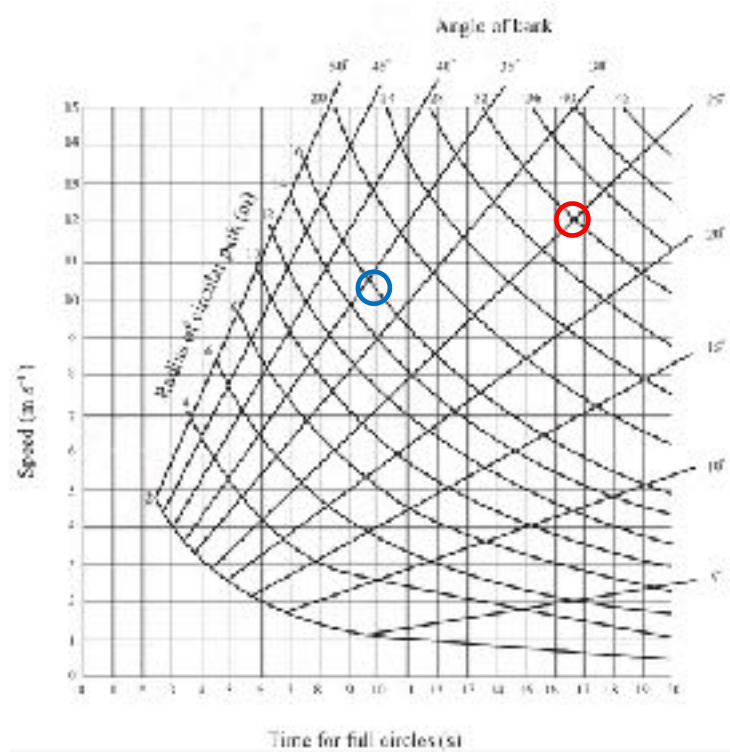
C 100 times as far as **P**

D 1000 times as far as **P**.

Since Intensity $I \propto \frac{1}{r^2}$, increasing distance by a factor of 10 decreases intensity by a factor 100, bringing **Q** back to **P**.

GAMSAT Question

Questions 67 and 68
Many birds are able to soar (fly without flapping their wings) by circling in the air. The air provides a lifting force on the outstretched wings of a soaring bird. The lifting force partly opposes the bird's weight and helps to keep it in the air.
For a soaring bird, the figure shows the relationships between its speed, the time it takes to complete one full circle of a horizontal circular path, the radius of its circular path, and its angle of bank.
The angle of bank is the angle between the line joining the tips of the outstretched wings of the bird and the horizontal plane containing the bird's path. When the bird banks, the lift acting on its wings is tilted from the vertical by the angle of bank. The lift has horizontal and vertical components, but only the vertical component of the lift is available to counter the bird's weight.
For example, as shown in the figure, if the speed is 4 ms $^{-1}$ and the time for one complete circle is 11 s, the circular path of a soaring bird has a radius of approximately 7 m and an angle of bank of approximately 13°.



67 A bird of mass 2 kg is soaring in a circular path. The speed of the bird is 12 m s $^{-1}$ and its angle of bank is 25°.
If the bird halves the radius of its circular path and increases its angle of bank by 10°, which of the following is the best estimate of the bird's speed in its new path?

A. 2 ms $^{-1}$
B. 4 m s $^{-1}$
C. 8 m s $^{-1}$
D. 10 m s $^{-1}$

The initial location of the bird on the graph is shown as Red circle. $\Rightarrow radius,\ r = 32\ m$.

E. Halving the radius puts $r = 16\ m$, increasing the bank angle makes $35°$ => shown by blue circle

Reading off the speed shown by the blue circle gives $10.5\ ms^{-1}$ (Ans D)

GAMSET Question 78 -78 continued...

68. A bird of mass 2.5 kg is soaring in a circular path with a radius of 10 m. The magnitude of the centripetal force acting on it is equal to the magnitude of its weight.

Centripetal force can be calculated from the following formula:

$$F = \frac{mv^2}{r}$$

Which of the following is the best estimate of the angle of bank at which the bird is soaring?

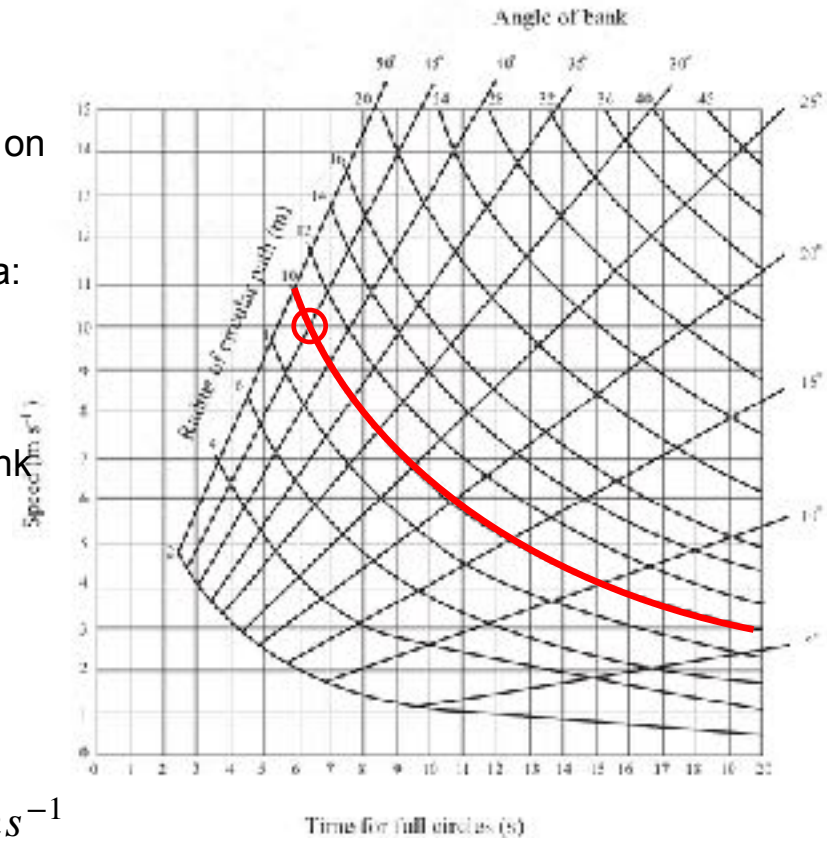
A 30°
B 35°
C 40°
D 45°

Red curve shows 10m locus. Need to find speed to determine bank angle.

Weight=centripetal force $\Rightarrow \frac{mv^2}{r} = mg$

$\Rightarrow v^2 = rg \Rightarrow v^2 = 10 \times 10 \Rightarrow v = 10\ ms^{-1}$

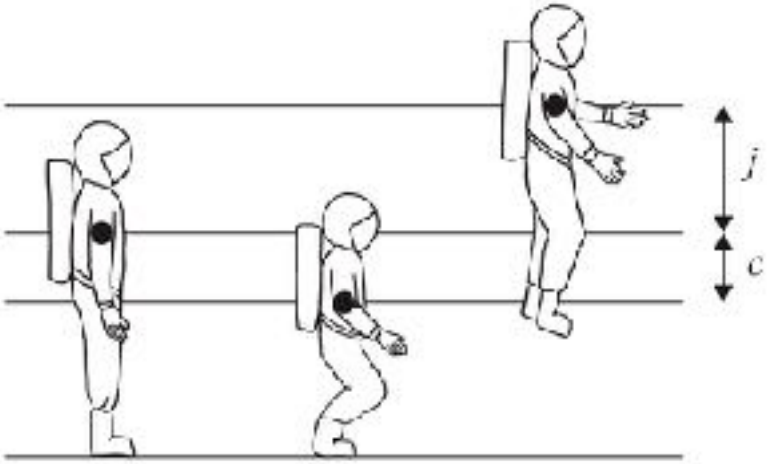
Shown by red circle reading off bank angle gives 45° (Ans D)



GAMSAT Question 57- 60

Questions **57 – 60**

A person jumps by first crouching, thus lowering their centre of mass by a distance, c, and then pushing downwards with constant force, F, to extend their legs. The jump raises their centre of mass from its original position by a distance, j.



Assume that the force needed for a person to raise their centre of mass by j is k times their weight:

$$F = kmg$$

Where m is mass, g is acceleration due to local gravity

The force of gravity on the surface of the Moon is one-sixth as strong as on the surface of Earth. It can be shown that, for the same magnitude F,

$$j_{Earth} = c\left(k_{Earth} - 1\right) \text{ and}$$

$$j_{Moon} = c\left(k_{Moon} - 1\right) = c(6k_{Earth} - 1)$$

Assume that m and c are the same for all jumps.

57 Consider jumps on Earth and the Moon.
A person will only leave the ground

A on Earth if $\boldsymbol{k}_{Earth} = 1$ and on the Moon if $\boldsymbol{k}_{Moon} = 1$.
B on Earth if $\boldsymbol{k}_{Earth} > 1$ and on the Moon if $\boldsymbol{k}_{Moon} > 1$.
C on Earth if $\boldsymbol{k}_{Earth} = 1$ but on the Moon only if $\boldsymbol{k}_{Moon} = 6$.
D on Earth if $\boldsymbol{k}_{Earth} > 1$ but on the Moon only if $\boldsymbol{k}_{Moon} > 6$.

Clearly only leave ground if $\left(\boldsymbol{k} - 1\right) > 0$

So $\boldsymbol{k} > 1$ for both Earth and Moon
(Ans B)

GAMSAT Question 58-59

Assume that the force needed for a person to raise their centre of mass by j is k times their weight:

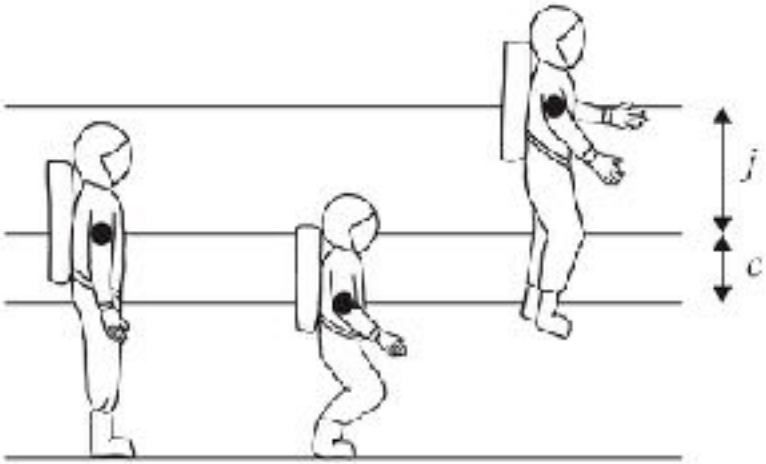
$F = kmg$

Where m is mass, g is acceleration due to local gravity

The force of gravity on the surface of the Moon is one-sixth as strong as on the surface of Earth. It can be shown that, for the same magnitude F,

$j_{Earth} = c\left(k_{Earth} - 1\right)$ and $j_{Moon} = c\left(k_{Moon} - 1\right) = c(6k_{Earth} - 1)$

Assume that m and c are the same for all jumps.



58 Suppose that a person jumps 11 times as high on the Moon as on Earth when using equal force. How large is the force exerted by the person jumping?

A the same as the person's weight on Earth
B the same as the person's weight on the Moon
C twice the person's weight on Earth
D twice the person's weight on the Moon

Force on Earth and Moon is the same $\Rightarrow k_E m g_E = k_M m g_M$

$\Rightarrow k_M = \dfrac{k_E g_E}{g_M} = 6k_E$, now $j_{Moon} = 11 \times j_{Earth}$ so

$c\left(k_M - 1\right) = 11 \times c\left(k_E - 1\right) \Rightarrow 6k_E - 1 = 11k_E - 11 \Rightarrow$

$5k_E = 10 \Rightarrow k_E = 2$

So force is twice weight on Earth (Ans C)

59 Consider a person who jumps on the Moon using a force that is six times their weight on the Moon. If the person uses equal force on Earth, they will:

A not leave the ground.
B jump as high as on the Moon.
C jump one-sixth as high as on the Moon.
D jump five-sixths as high as on the Moon.

$$F = 6mg_M = \frac{6mg_E}{6} = mg_E \Rightarrow k_E = 1$$

Subbing into equation for $j_{Earth} = c\left(k_E - 1\right) = 0$

So doesn't leave the ground on Earth. (Ans **A**)

GAMSAT Question 60 continued …

Assume that the force needed for a person to raise their centre of mass by j is k times their weight:

$F = kmg$

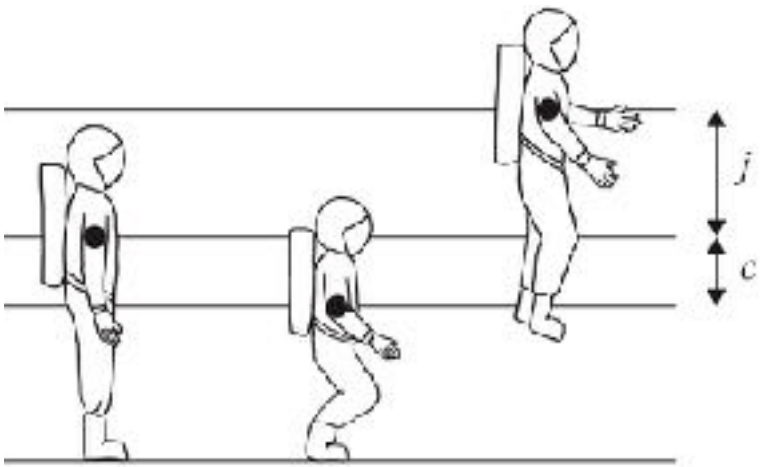
Where m is mass, g is acceleration due to local gravity

The force of gravity on the surface of the Moon is one-sixth as strong as on the surface of Earth. It can be shown that, for the same magnitude F,

$j_{Earth} = c\left(k_{Earth} - 1\right)$

and $j_{Moon} = c\left(k_{Moon} - 1\right) = c(6k_{Earth} - 1)$

Assume that m and c are the same for all jumps.



60 Consider a planet with one-third the gravity of the Earth.
For jumps using equal force on the other planet and on Earth, which of the following is correct?

A $\dfrac{j_{Moon}}{j_{Planet}} > \dfrac{j_{Moon}}{j_{Earth}}$

B $\dfrac{j_{Moon}}{j_{Planet}} < \dfrac{j_{Moon}}{j_{Earth}}$

C $\dfrac{j_{Moon}}{j_{Planet}} = 3\dfrac{j_{Moon}}{j_{Earth}}$

D $\dfrac{j_{Moon}}{j_{Planet}} = 2$

We have: $j_{Planet} = c\left(k_P - 1\right)$

Ratio $\dfrac{g_M}{g_P} = \dfrac{g_M}{g_E}\dfrac{g_E}{g_P} = \dfrac{1}{6} \times 3 = \dfrac{1}{2}$ so:

$j_{Moon} = c\left(k_M - 1\right) = c\left(6k_E - 1\right) = c(2k_P - 1)$

So B is correct

GAMSAT Questions 37-39

Questions 37 -39

A lens is a piece of transparent material that is generally curved on both surfaces. If the thickness of the lens is small compared with its focal length (the distance at which it focuses light) then the thin lens formula can be used to determine the overall focal length of the lens. The thin lens formula is:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Where:

f is the focal length of the lens,

n is the refractive index of the material of the lens,

R_1 is the radius of curvature of the surface nearest to the light source,

R_2 is the radius of curvature of the surface furthest from the light source

The sign convention for focal length is that f is positive for a converging lens (parallel rays of light enter the lens and are focused to a point on the other side) and negative for a diverging lens (parallel rays of light enter the lens and spread out, appearing to come from a point).

The sign convention for the radii of curvature is as follows:

	Convex Surface	**Concave surface**
R_1	positive	negative
R_2	negative	positive

37 The focal length of biconvex lens that has curvature radii of 60cm on both sides and is made of a material with a refractive index of 1.75 is: (Ans B)

A 20 cm

B 40 cm

C 50 cm

D 67 cm

Subbing into the equation with the sign conventions gives:

$$\frac{1}{f} = (1.75-1)\left(\frac{1}{0.6} - \left(-\frac{1}{0.6}\right)\right) = 0.75\left(\frac{1}{0.3}\right) = 2.5$$

$\Rightarrow f = 40\ cm$ (Ans **B**)

GAMSAT Questions 38 - 39

38 Consider a biconvex lens for which the radius of curvature of one side is quite different to the radius of curvature of the other side.
What would be the effect of reversing this lens?

A Its focal length would be the inverse of the original focal length.
B If the lens was originally a converging lens, it would now be a diverging lens
C There would be no change; the lens would refract light in the same way.
D The values of n, R_1 and R_2 need to be known in order to determine if there is a change

Reversing the direction of light through a lens does not change it's focal length.
Ans **C**

39 Consider a symmetrical biconvex lens that has a focal length as long as its radius of curvature.
What is the refractive index of the material used to make this lens?

A 1.33
B 1.50
C 1.75
D 2.00

Subbing into the formula with the sign conventions gives:

$$\frac{1}{f} = (n-1)\left(\frac{1}{f} - \left(-\frac{1}{f}\right)\right) = \frac{(n-1)2}{f}$$

$$\Rightarrow n - 1 = \frac{f}{f} \times \frac{1}{2} \Rightarrow n = 1 + \frac{1}{2} = 1.5$$

Ans **B**

GAMSAT Question 21

The most common domestic voltages used around the world are 240 V and 110 V. In general, the risk to a person
who makes contact with an electrical source is directly proportional to the size of the current.

The table gives the typical electrical resistance of the human body for two difference contact voltages.

Voltage (V)	Resistance (Ω)
110	3000
240	2200

21 Assuming all other factors are equal, which of the following is the closest estimate of the ratio of the risk to a person from a 240 V source compared to a 110 V source?

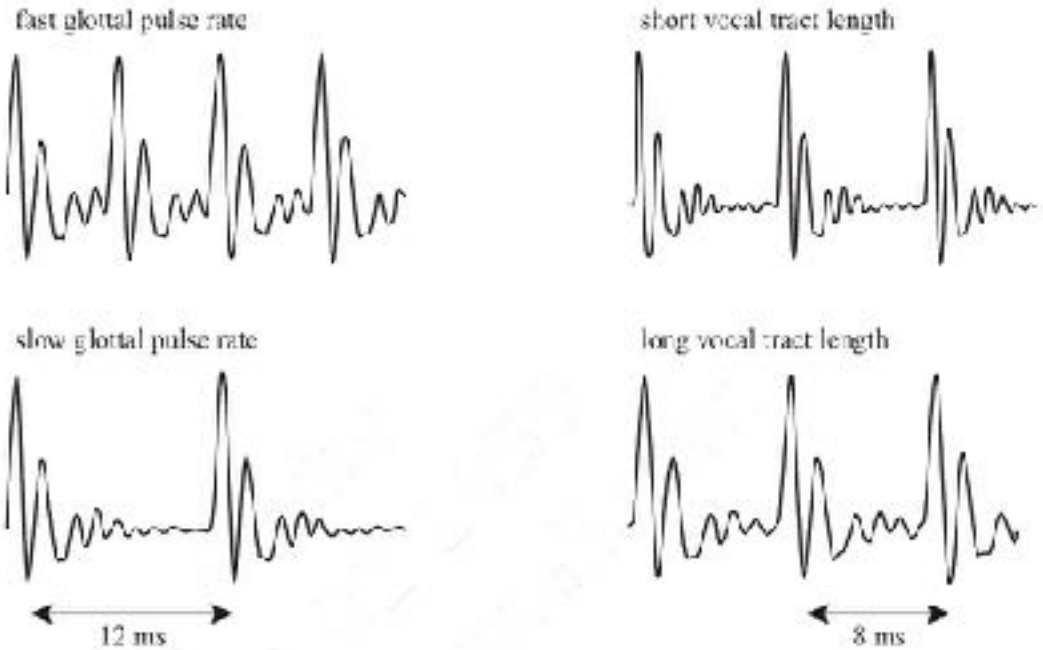
A 2 : 3
B 2 : 1
C 3 : 1
D 4 : 3

$$I_{240} = \frac{240}{2200} = 0.109, \; I_{110} = \frac{110}{3000} = 0.0367$$

Answer **C**

GAMSAT Questions 5 - 7

The four graphs below show the same vowel sound (fundamental and resulting resonances) when produced through the vocal tract under different conditions. The distance between two peaks of the same amplitude represents one pulse. The glottal pulse rate is the rate of opening and closing of the vocal folds.



$f = \frac{1}{T}$ where f is frequency (Hz) and T is period (s)

All the graphs are drawn to the same scale.

5 What is the frequency of the sound produced through the short vocal tract length?

A 0.125 Hz
B 1.25 Hz
C 12.5 Hz
D 125 Hz

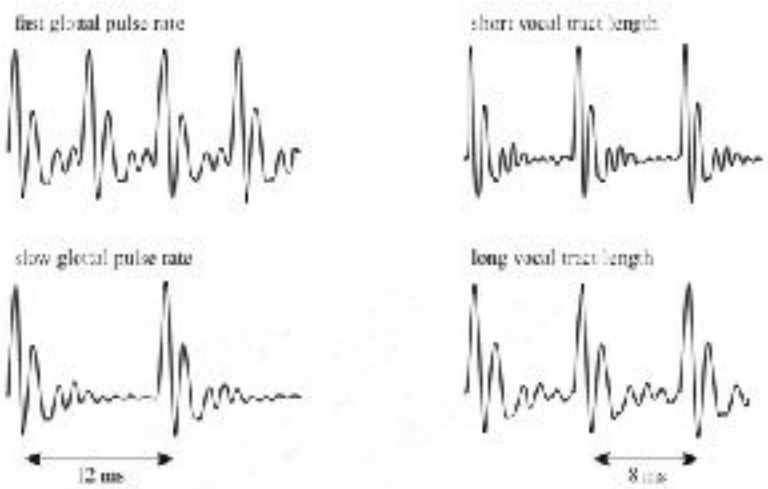
Period $T = 8ms \Rightarrow f = \frac{1}{T} = \frac{1}{0.008} = 125\ Hz$

Answer **D**

GAMSAT Questions 6 - 7

6 Compared with a long vocal tract length, for a short vocal tract:

A the period of the sound is shorter.
B the resonance decays much more slowly.
C the resonance decays much more quickly.
D the frequency of the sound emitted is higher.



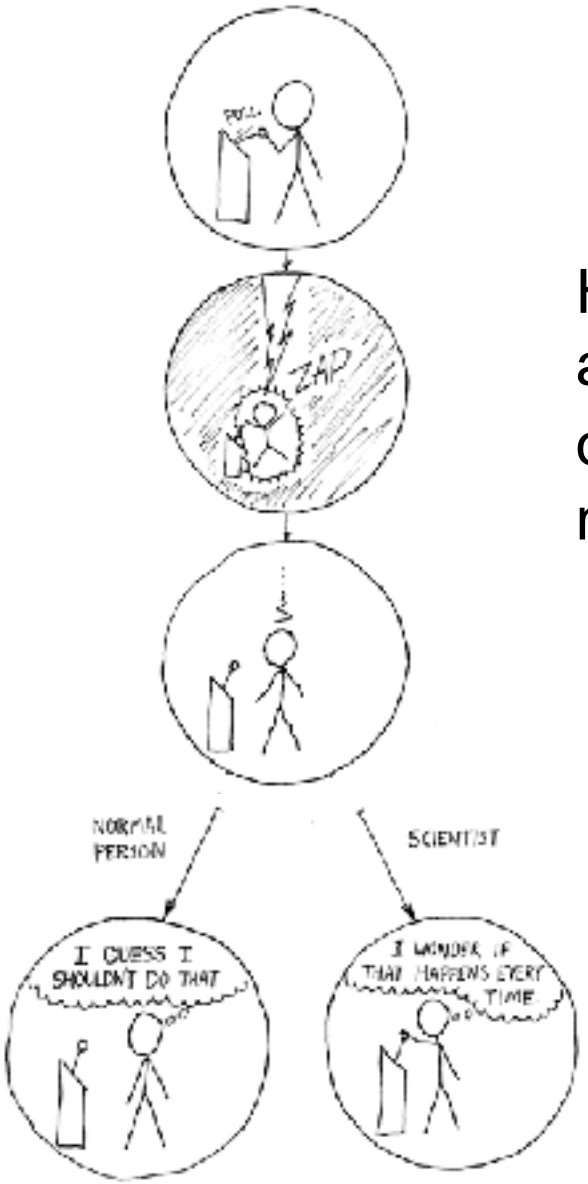
The short vocal tract length shows oscillations diminishing to close to zero between spikes hence the resonance decays quicker than the long vocal tract. Answer **C**

7 A longer vocal tract length and a faster glottal pulse rate result in a period

A decrease and resonance lasts longer.
B increase and resonance lasts longer.
C decrease and resonance reduces more quickly.
D increase and resonance reduces more quickly.

Reading off the graphs the long vocal length has shorter period, The fast glottal pulse rate resonance is longer so answer is **A**

Finally....



How could you choose avoiding a little pain over **understanding** a magic lightening machine?

https://xkcd.com/242/

33